CIE A Level Chemistry

Your notes

9.2 Periodicity of Chemical Properties of the Elements in Period 3

Contents

Reactions of the Period 3 Elements

Your notes

Reactions of the Period 3 Elements

Reactions with oxygen & chlorine

Reaction of Period 3 elements with oxygen table

	Chemical equation	Reaction conditions	Reaction	Flame	Product
Na	$4Na (s) + O_2 (g) \rightarrow 2Na_2O (s)$	Heat	Vigorous	Bright yellow	White solid
Mg	$2Mg (s) + O_2 (g) \rightarrow 2MgO (s)$	Heat	Vigorous	Bright white	White solid
Al	$4Al (s) + 3O_2 (g) \rightarrow 2Al_2O_3 (s)$	Powdered Al	Fast	Bright white	White powder
Si	$Si (s) + O_2 (g) \rightarrow SiO_2 (s)$	Powdered Si Heat strongly	Slow	Bright white sparkles	White powder
P	$4P (s) + 5O_2 (g) \rightarrow P_4O_{10} (s)$	Heat	Vigorous	Yellow or white	White clouds
S	$S (s) + O_2 (g) \rightarrow SO_2 (g)$	Powdered S Heat	Gentle	Blue	Toxic fumes

Reaction of Period 3 elements with chlorine table

	Chemical equation	Reaction conditions	Reaction
Na	$2Na (s) + Cl_2 (g) \rightarrow 2NaCl (s)$	Heat	Vigorous
Mg	$Mg (s) + Cl_2 (g) \rightarrow MgCl_2 (s)$	Heat	Vigorous

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Al	$2Al\ (s) + 3Cl_2\ (g) \rightarrow Al_2Cl_6\ (s)$	Heat	Vigorous
Si	$Si\ (s) + 2Cl_2\ (g) \rightarrow SiCl_4\ (l)$	Heat	Slow
P	$2P\ (s) + 5Cl_2\ (g) \rightarrow 2PCl_5\ (l)$	Heat Excess chlorine	Slow

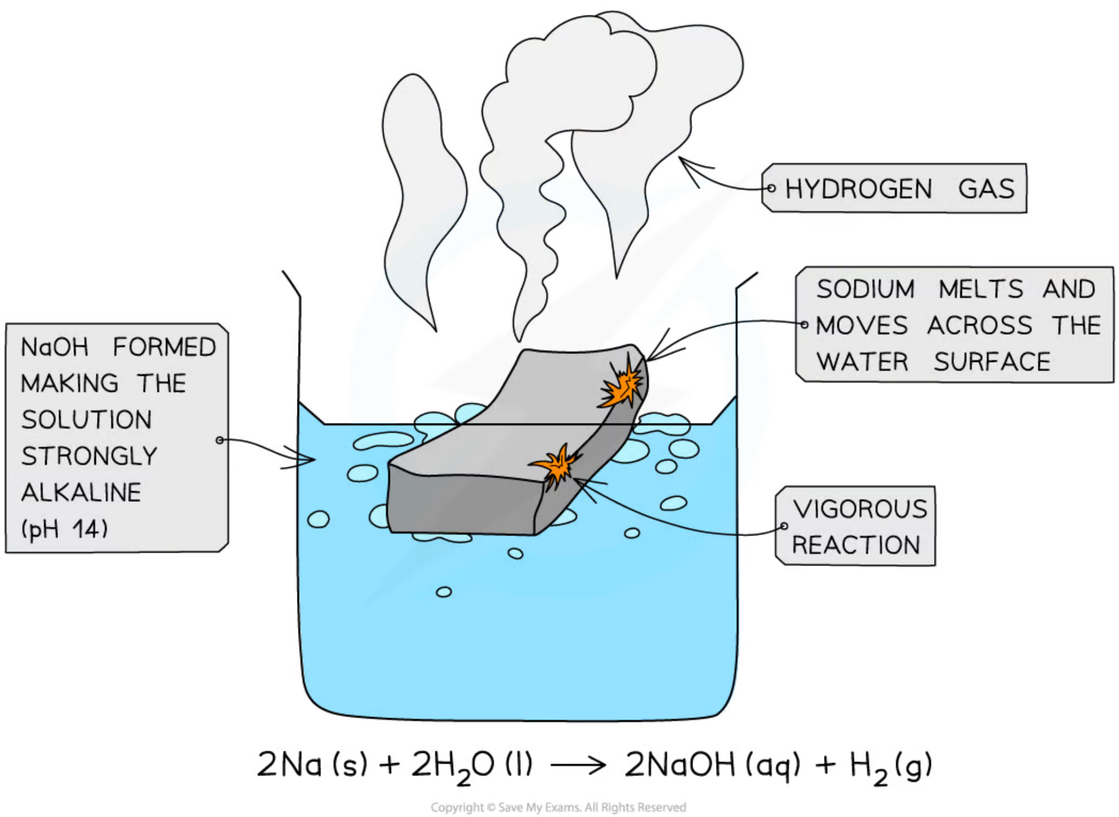
Reaction of sodium with water

- Sodium reacts **vigorously** with **cold water**:

$$2Na\ (s) + 2H_2O\ (l) \rightarrow 2NaOH\ (aq) + H_2\ (g)$$

- The sodium **melts** into a ball and moves across the water surface until it disappears
- **Hydrogen gas** is given off
- The solution formed is **strongly alkaline** (pH 14) due to the **sodium hydroxide** which is formed

The reaction of sodium with cold water



Sodium fizzes while vigorously reacting with water to form sodium hydroxide

Reaction of magnesium with water

- Magnesium reacts **extremely slowly** with cold water:

$$Mg\ (s) + 2H_2O\ (l) \rightarrow Mg(OH)_2\ (aq) + H_2\ (g)$$

- The solution formed is **weakly alkaline** (pH 11) as the formed **magnesium hydroxide** is only **slightly soluble**
- When magnesium is **heated**, it reacts **vigorously** with steam (water) to make magnesium oxide and hydrogen gas:

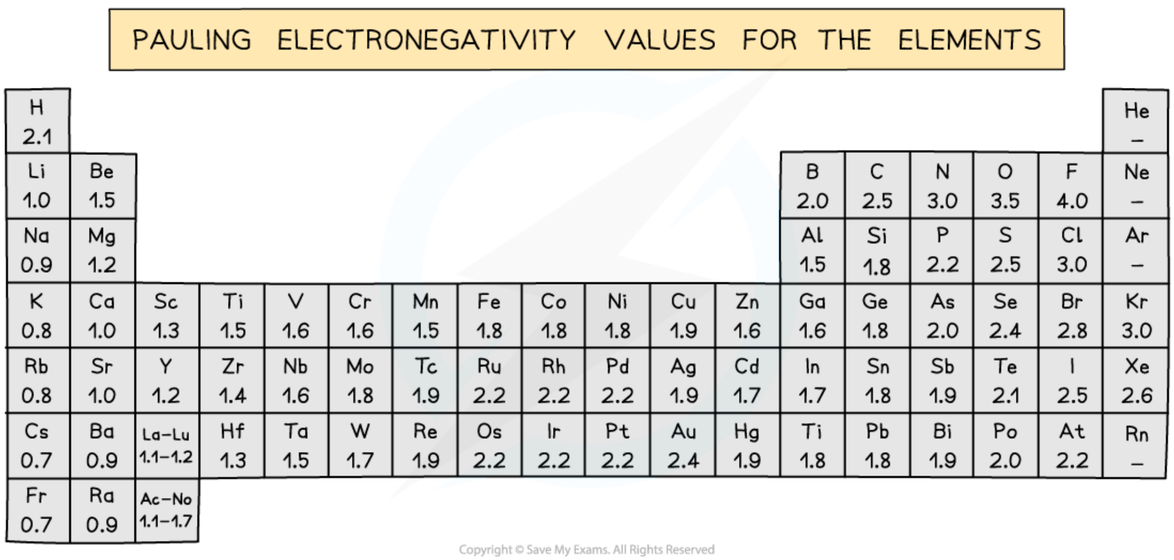
$$Mg\ (s) + H_2O\ (g) \rightarrow MgO\ (s) + H_2\ (g)$$

Oxidation Number of Period 3 Oxides & Chlorides

Your notes

- Chlorine is more **electronegative** than the other Period 3 elements
  - Therefore, the other Period 3 elements will generally have **positive oxidation states** in their chlorides, while the chlorine has a **negative oxidation state** of -1
- Oxygen is more **electronegative** than any of the Period 3 elements
  - Therefore, the Period 3 elements will have **positive oxidation states** in their oxides, while the oxygen has a **negative oxidation state** of -2

The Pauling scale of electronegativity



The Pauling scale shows that chlorine has a higher electronegativity than the other Period 3 elements and oxygen has a higher electronegativity than all of the Period 3 elements

Oxidation numbers of Period 3 elements in their oxides

- The normal Oxidation Number Rules apply to the Period 3 oxides

Table of formulae of the Period 3 oxides & the elements oxidation numbers

Period 3 element	Na	Mg	Al	Si	P	S
Formula of oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2 SO_3
Oxidation number of the Period 3 element	+1	+2	+3	+4	+5	+4 +6

Your notes

Exam Tip

- SiO_2 is not written in the specification as expected knowledge but you should be able to apply the oxidation number rules to this and other chemicals

Worked example

Deducing oxidation numbers of Period 3 elements in their oxides

State the oxidation number of the bold atoms in these compounds or ions.

1. **Na**$_2$O
2. **Al**$_2$O$_3$
3. **P**$_4$O$_{10}$

Answer

1. **Na**$_2$O
    - 1 O atom = –2
    - The overall charge of the compound = 0
    - 2 Na atoms = +2
    - Oxidation number of 1 Na atom = (+2) / 2 = +1
2. **Al**$_2$O$_3$
    - 3 O atoms = 3 x (–2) = –6
    - The overall charge of the compound = 0
    - 2 Al atoms = +6
    - Oxidation number of 1 Al atom = (+6) / 2 = +3
3. **P**$_4$O$_{10}$
    - 10 O atoms = 10 x (–2) = –20
    - The overall charge of the compound = 0
    - 4 P atoms = +20
    - Oxidation number of 1 Al atom = (+20) / 4 = +5

Oxidation numbers of Period 3 elements in their chlorides

- The normal Oxidation Number Rules apply to the Period 3 chlorides

Table of formulae of the Period 3 chlorides & the elements oxidation numbers

Period 3 element	Na	Mg	Al	Si	P
Formula of chloride	$NaCl$	$MgCl_2$	$AlCl_3$	$SiCl_4$	PCl_5

Oxidation number of the Period 3 element	+1	+2	+3	+4	+5

Your notes

Worked example

Deducing oxidation numbers of Period 3 elements in their chlorides

State the oxidation number of the bold atoms in these compounds or ions.

1. **Na**$_2$O
2. **Al**$_2$O$_3$
3. **P**$_4$O$_{10}$

Answer

1. **Mg**Cl$_2$
   - 2 Cl atoms = 2 x (–1) = –2
   - The overall charge of the compound = 0
   - Oxidation number of 1 Mg atom = +2
   - Or, the oxidation number of a Group 2 metal is +2
2. **Al**Cl$_3$
   - 3 Cl atoms = 3 x (–1) = –3
   - The overall charge of the compound = 0
   - Oxidation number of 1 Al atom = +3
3. **P**Cl$_5$
   - 5 Cl atoms = 5 x (–1) = –5
   - The overall charge of the compound = 0
   - Oxidation number of 1 P atom = +5

Reaction of Period 3 Oxides & Water

Your notes

- Not all Period 3 oxides **react** with or are **soluble** in water

Reaction of Period 3 oxides with water table

Oxide	Chemical equation	pH	Comments
Na_2O	Na_2O (s) + H_2O (l) → 2NaOH (aq)	12 - 14 (Strongly alkaline)	-
MgO	MgO (s) + H_2O (l) → $Mg(OH)_2$ (aq)	8 - 10 (Weakly alkaline)	-
Al_2O_3	No reaction	-	Al_2O_3 is insoluble in water
SiO_2	No reaction	-	SiO_2 is insoluble in water
P_4O_{10}	P_4O_{10} (s) + 6H_2O (l) → 4H_3PO_4 (aq)	3 - 4 (Weakly acidic)	Vigorous reaction
SO_2 **SO_3**	SO_2 (s) + H_2O (l) → H_2SO_3 (aq) SO_3 (s) + H_2O (l) → H_2SO_4 (aq)	1 - 2 (Strongly acidic)	-

Exam Tip

- Since aluminium oxide does not react or dissolve in water, the oxide layer protects the aluminium metal from corrosion
- The reaction of silicon(IV) oxide is not required knowledge

Your notes

Acid/Base Behaviour of the Period 3 Oxides & Hydroxides

Acid / Base Behaviour of Period 3 Oxides & Hydroxides

Period 3 oxides

- Aluminium oxide is **amphoteric** which means that it can act both as a base (and react with an acid such as HCl) and an acid (and react with a base such as NaOH)

Acidic and basic nature of Period 3 oxides table

Period 3 oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2 SO_3
Acid / base nature	Basic	Basic	Amphoteric	Acidic	Acidic	Acidic

Reaction of Period 3 oxides with acid / base table

Oxide	Chemical equation	Comment
Na_2O	$Na_2O\ (s) + 2HCl\ (aq) \rightarrow 2NaCl\ (aq) + H_2O\ (l)$	
MgO	$MgO\ (s) + 2HCl\ (aq) \rightarrow MgCl_2\ (aq) + H_2O\ (l)$	Used in indigestion remedies by neutralising excess acid in the stomach
Al_2O_3	$Al_2O_3 (s) + 3H_2SO_4\ (aq) \rightarrow Al_2(SO_4)_3 (aq) + 3H_2O\ (l)$ $Al_2O_3 (s) + 2NaOH\ (aq) + 3H_2O\ (l) \rightarrow 2NaAl(OH)_4\ (aq)$	Reacts with acid to form salt and water Reacts with hot, concentrated alkali to form salt
SiO_2	$SiO_2\ (s) + 2NaOH\ (aq) \rightarrow Na_2SiO_3 (aq) + H_2O\ (l)$	
P_4O_{10}	$P_4O_{10}\ (s) + 12NaOH\ (aq) \rightarrow 4Na_3PO_4\ (aq) + 6H_2O\ (l)$	
SO_2	$SO_2\ (g) + 2NaOH\ (aq) \rightarrow Na_2SO_3 (aq) + H_2O\ (l)$	

SO_3	$SO_3(g) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + H_2O(l)$	

Your notes

- The acidic and basic nature of the Period 3 elements can be explained by looking at their **structure**, **bonding** and the Period 3 elements' **electronegativity**

Structure & bonding of the Period 3 oxides table

Period 3 oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2 SO_3
Relative melting point	High	High	Very high	Very high	Low	Low
Chemical bonding	Ionic	Ionic	Ionic (with some covalent character)	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Giant ionic	Giant covalent	Simple molecular	Simple molecular

Electronegativity of Period 3 elements table

Period 3 element	Na	Mg	Al	Si	P	S	Cl
Electronegativity	0.9	1.2	1.5	1.8	2.1	2.5	3.0

- Oxygen has an electronegativity value of 3.5
- Therefore, the difference in electronegativity between oxygen and Na, Mg and Al is the largest
- So, electrons will be **transferred** to oxygen when forming oxides giving the oxide an **ionic binding**
- The oxides of Si, P and S will **share** the electrons with the oxygen to form **covalently bonded** oxides
- The **giant ionic** and **giant covalent** structured oxides will have **high melting points** as it is difficult to break the structures apart

The structure of some Period 3 oxides

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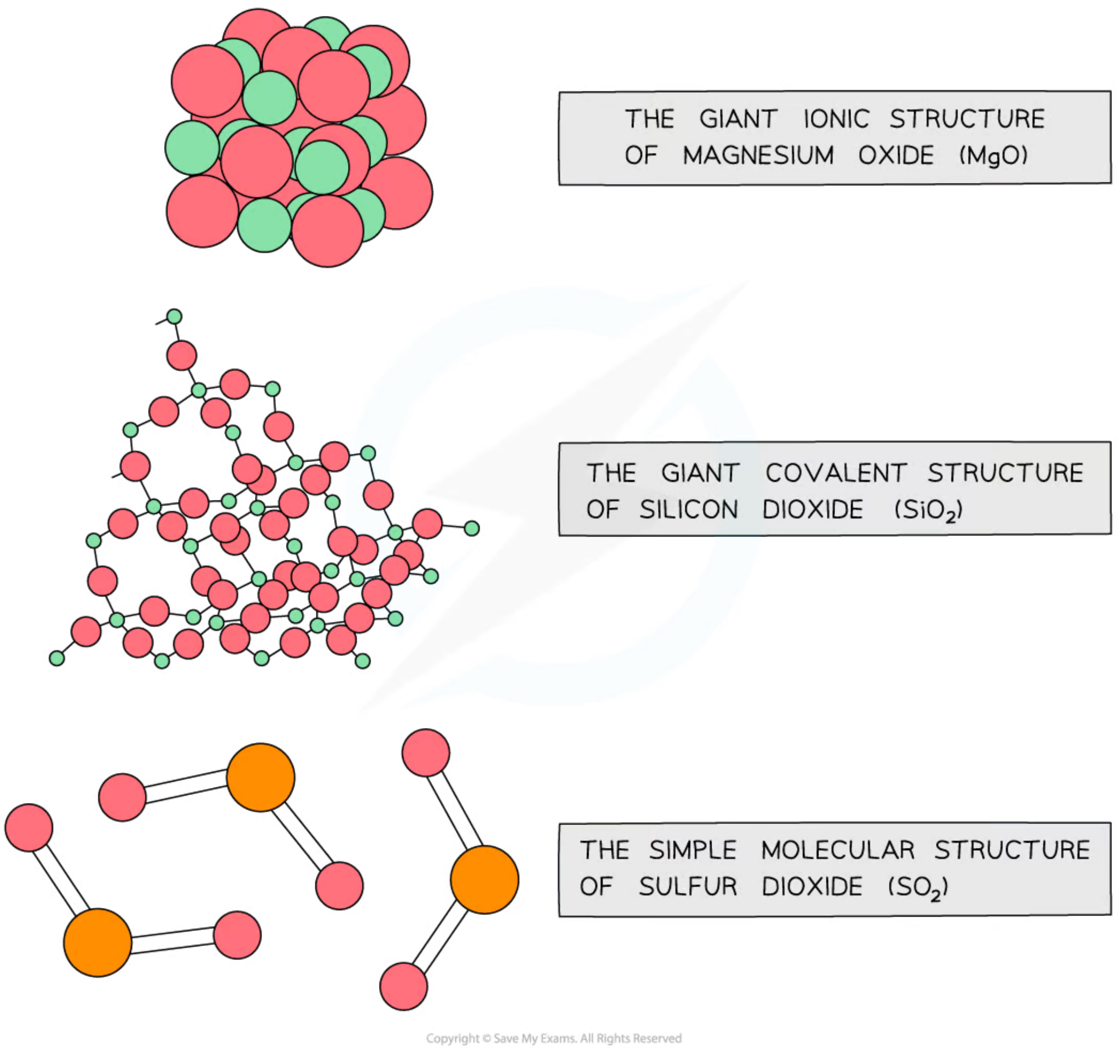


Magnesium oxide is giant ionic, silicon dioxide is giant covalent and sulfur dioxide is simple molecular.

- The oxides of **Na and Mg** which show purely **ionic bonding** produce **alkaline** solutions with water as their **oxide** ions (O^{2-}) become **hydroxide** ions (OH^-):

$$O^{2-}\ (aq) + H_2O\ (l) \rightarrow 2OH^-\ (aq)$$

- The oxides of **P and S** which show purely **covalent bonding** produce **acidic** solutions with water because when these oxides react with water, they form an acid which donates H^+ ions to water
  - Eg. SO_3 reacts with water as follows:

$$SO_3\ (g) + H_2O\ (l) \rightarrow H_2SO_4\ (aq)$$

  - The H_2SO_4 is an acid which will donate a H^+ to water:

$$H_2SO_4\ (aq) + H_2O\ (l) \rightarrow H_3O^+\ (aq) + HSO_4^-\ (aq)$$

- **Al** and **Si** are insoluble and when they react with **hot, concentrated alkaline solution** they act as a base and form a salt
  - This behaviour is very typical of a **covalently bonded oxide**
- **Al** can also react with **acidic solutions** to form a salt and water
  - This behaviour is very typical of an **ionic bonded metal oxide**
- This behaviour of **Al** proves that the chemical bonding in aluminium oxide is not purely **ionic** nor **covalent: it is amphoteric**

Period 3 hydroxides

- NaOH is a strong base and will react with acids to form a salt and water:

$$NaOH\ (aq) + HCl\ (aq) \rightarrow NaCl\ (aq) + H_2O\ (l)$$

- $Mg(OH)_2$ is also a basic compound which is often used in indigestion remedies by neutralising the excess acid in the stomach to relieve pain:

$$Mg(OH)_2\ (s) + 2HCl\ (aq) \rightarrow MgCl_2\ (aq) + 2H_2O\ (l)$$

- $Al(OH)_3$ is **amphoteric** and can act both as an acid and base:

$$Al(OH)_3\ (s) + 3HCl\ (aq) \rightarrow AlCl_3\ (s) + 3H_2O\ (l)$$

$$Al(OH)_3\ (s) + NaOH\ (aq) \rightarrow NaAl(OH)_4\ (aq)$$

Exam Tip

- Electronegativity is the power of an element to draw the electrons towards itself in a covalent bond.
- For example, in Na_2O the oxygen will draw the electrons more strongly towards itself than sodium does.

Your notes

Period 3 Chlorides

Your notes

Reaction of Period 3 Chlorides & Water

- Chlorides of Period 3 elements show characteristic behaviour when added to water which can be explained by looking at their **chemical bonding** and **structure**

Chemical bonding & structure of Period 3 chlorides table

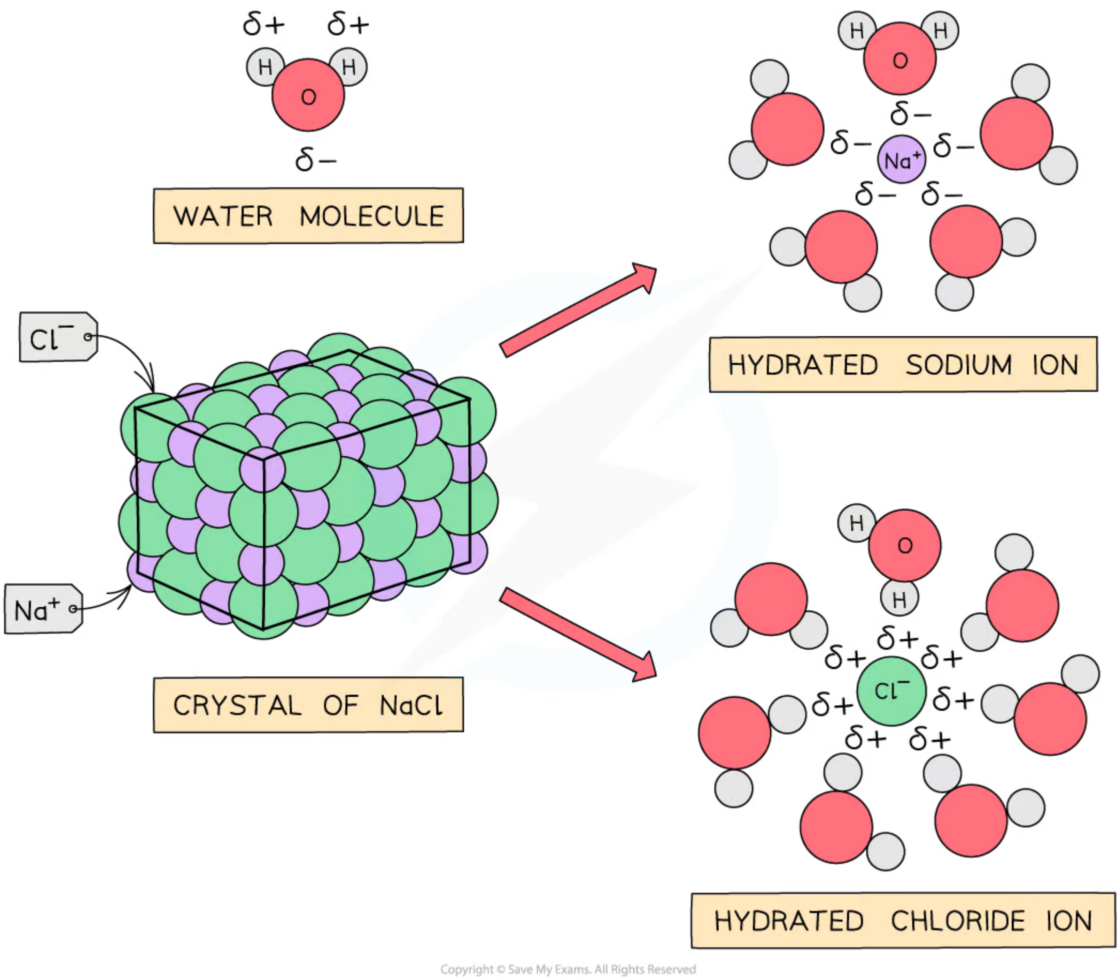
Period 3 chloride	NaCl	$MgCl_2$	Al_2Cl_6	$SiCl_4$	PCl_5	SCl_2
Chemical bonding	Ionic	Ionic	Covalent	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Simple molecular	Simple molecular	Simple molecular	Simple molecular
Observations	White solids dissolve to form colourless solutions		Chlorides react with water giving off white fumes of hydrogen chloride gas			
pH of solution formed	7.0	6.5	3.0	2.0	2.0	2.0

Sodium & magnesium chloride

- NaCl and $MgCl_2$ do not react with water as the **polar** water molecules are attracted to the ions **dissolving** the chlorides and breaking down the **giant ionic structures**: the metal and chloride ions become hydrated ions

How the giant ionic structure of NaCl and $MgCl_2$ break down in water

Your notes



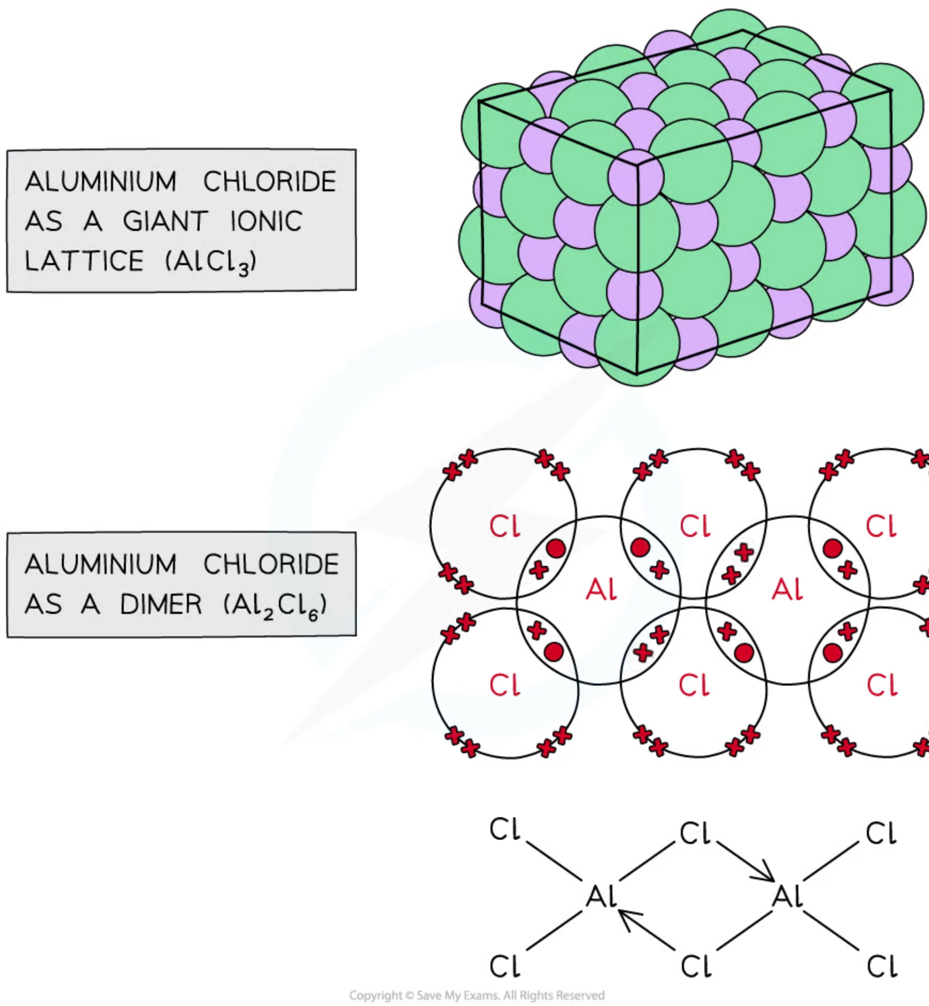
The diagram shows water molecules breaking down the giant ionic structure of NaCl and $MgCl_2$ to form hydrated ions

Aluminium chloride

- Aluminium chloride exists in two forms:
  - $AlCl_3$ is a **giant lattice** with **ionic** bonds
  - Al_2Cl_6 is a **dimer** with **covalent** bonds

The two forms of aluminium chloride

Your notes

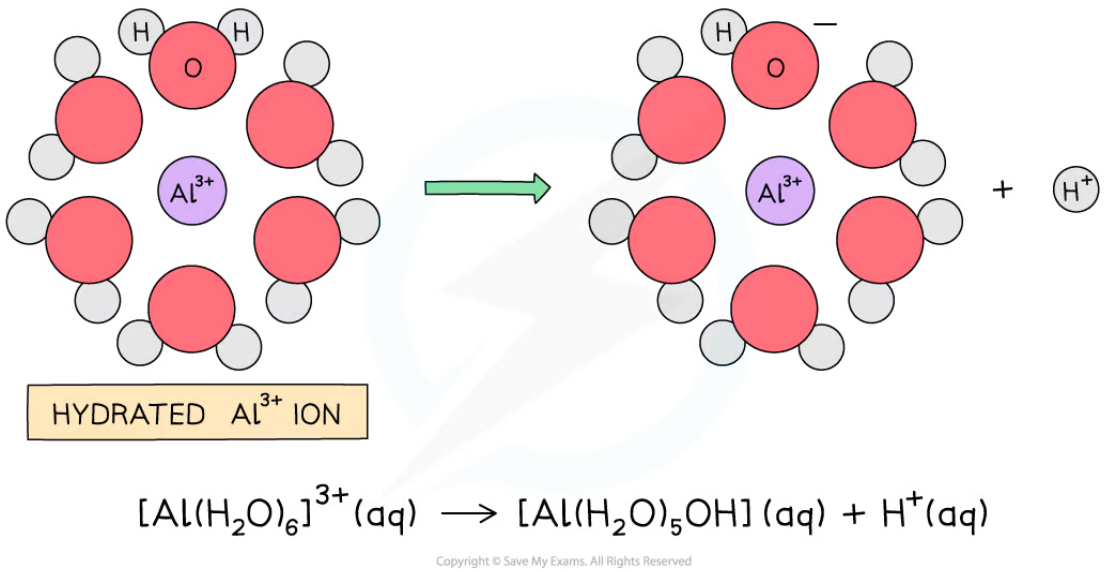


Aluminium chloride exists as a giant ionic lattice or a covalent dimer

- When water is added to aluminium chloride the dimers are broken down and Al^{3+} and Cl^- ions enter the solution
- The highly charged Al^{3+} ion becomes hydrated and causes a water molecule that is bonded to the Al^{3+} to lose an H^+ ion which turns the solution **acidic**
- The H^+ and the Cl^- form **hydrogen chloride gas** which is given off as **white fumes**

How the Al^{3+} makes an acidic solution

Your notes



The hydrated aluminium causes a water molecule to lose a H^+ ion turning the solution acidic

Silicon chloride

- $SiCl_4$ is **hydrolysed** in water, releasing **white fumes of hydrogen chloride gas** in a **rapid** reaction

$$SiCl_4\ (l) + 2H_2O\ (l) \rightarrow SiO_2\ (s) + 4HCl\ (g)$$

- The SiO_2 is seen as a **white precipitate** and some of the **hydrogen chloride gas** produced dissolves in water to form an **acidic** solution

Phosphorus(V) chloride

- PCl_5 also gets **hydrolysed** in water

$$PCl_5\ (s) + 4H_2O\ (l) \rightarrow H_3PO_4\ (aq) + 5HCl\ (g)$$

- Both H_3PO_4 and dissolved HCl are highly **acidic**

Your notes

Electronegativity & Bonding of the Period 3 Elements

Trends in Period 3 Electronegativity & Bonding

Electronegativity

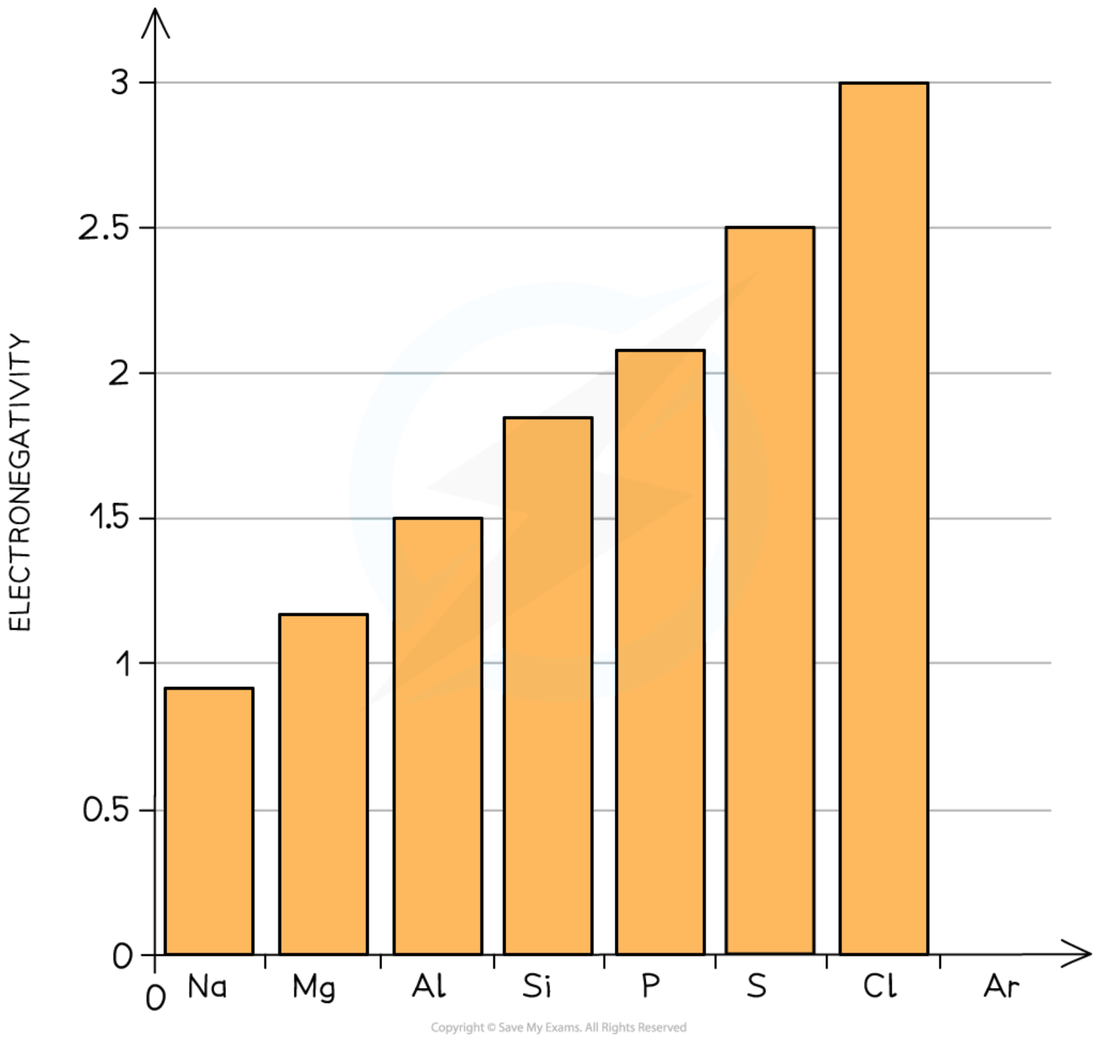
- **Electronegativity** is the power of an element to draw the electrons towards itself in a covalent bond
- Going **across** the period, the **electronegativity** of the elements increases

Electronegativity across Period 3 table

Period 3 element	Na	Mg	Al	Si	P	S	Cl	Ar
Electronegativity	0.9	1.2	1.5	1.8	2.1	2.5	3.0	-

Graph of the electronegativity of the Period 3 elements

Your notes



Electronegativity of the Period 3 elements increases from Na to Cl

- As the **atomic number** increases going across the period, there is an **increase** in **nuclear charge**
- Across the period, there is an increase in the number of **valence electrons** however the **shielding** is still the same as each extra electron enters the same shell
- As a result of this, electrons will be more strongly attracted to the nucleus causing an increase in electronegativity across the period

Bonding & structure of Period 3 elements table

Period 3 element	Na	Mg	Al	Si	P	S	Cl	Ar

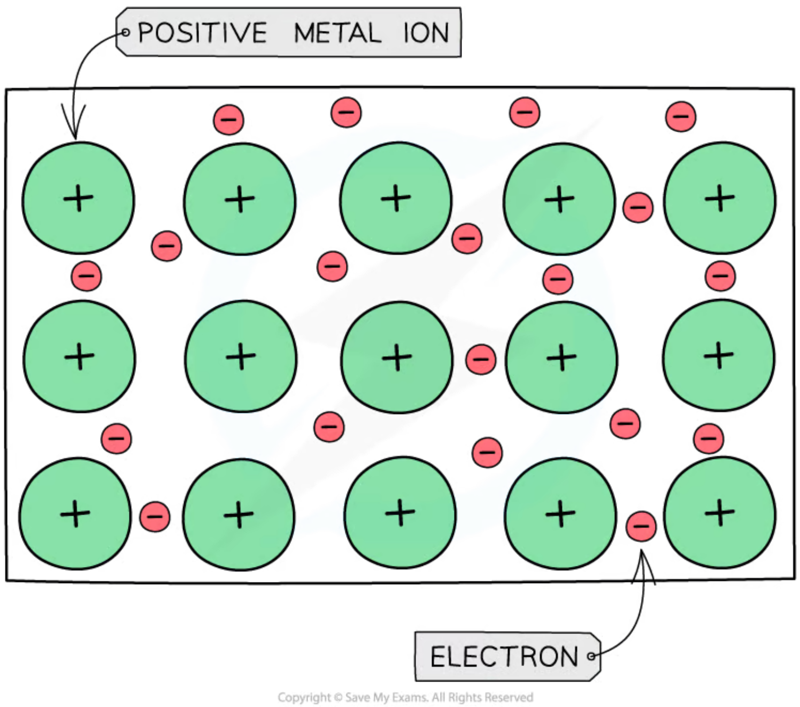
Bonding	Metallic	Metallic	Metallic	Covalent	Covalent	Covalent	Covalent	-
Structure	Giant metallic	Giant metallic	Giant metallic	Giant molecular	Simple molecular	Simple molecular	Simple molecular	Simple molecular

- The table shows that going from Al to S the bonding changes from **metallic** to **covalent** and the structure changes from **giant** to **simple** structure
- **Na**, **Mg** and **Al** are metallic elements which form positive ions arranged in a **giant lattice** in which the ions are held together by a 'sea' of delocalised electrons around them
- Since Al donates three electrons into the sea of delocalised electrons to form an ion with +3 charge, the **electrostatic forces** between the electrons and the aluminium ion will be very strong
- The electrons in the 'sea' of delocalised electrons are those from the **valence shell** of the atoms
- **Na** will donate one electron into the 'sea' of delocalised electrons, **Mg** will donate two and **Al** three electrons
- As a result of this, the metallic bonding in **Al** is stronger than in **Na**
- This is because the electrostatic forces between a **3+ ion** and the larger number of negatively charged delocalised electrons are much larger compared to a **1+ ion** and the smaller number of delocalised electrons in Na
- Since there are more electrons in a metallic lattice of aluminium compared to sodium and magnesium, aluminium is a better electrical conductor

A giant metallic lattice

Your notes

Your notes

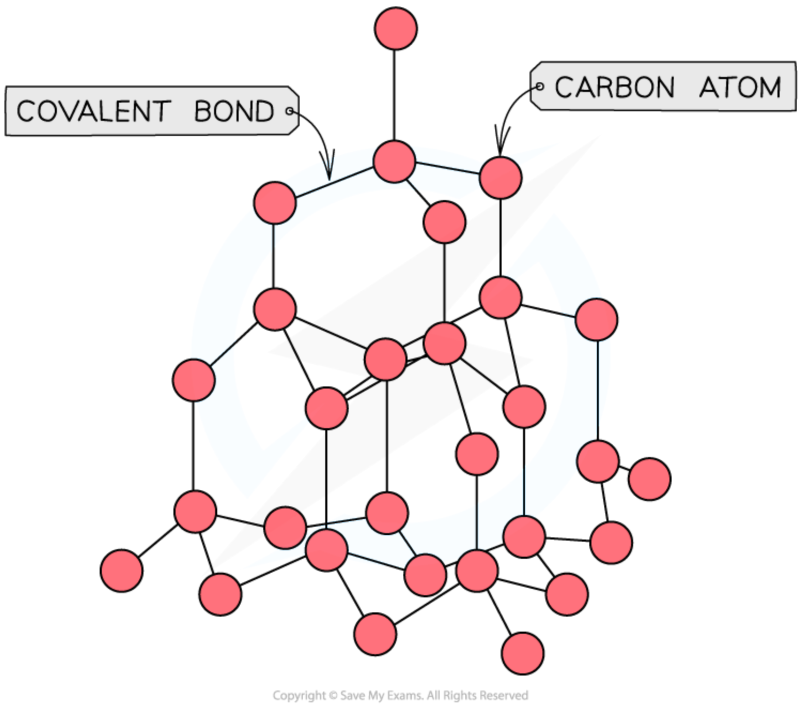


Metal cations form a giant lattice held together by electrons that can freely move around

- Si is a non-metallic element and has a **giant molecular** structure in which each Si atom is held to its neighbouring Si atoms by **strong covalent bonds**
- There are no delocalised electrons in the structure of Si which is why silicon cannot conduct electricity and is classified as a **metalloid**

The giant molecular structure of silicon

Your notes

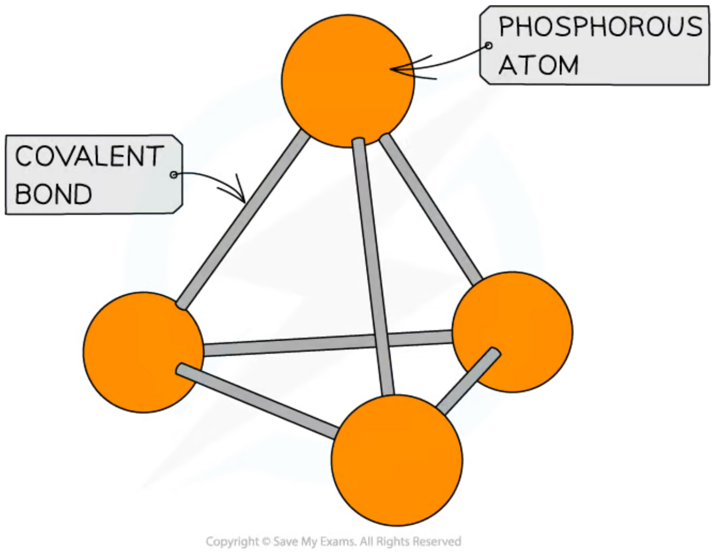


The diagram shows the giant molecular structure of silicon where silicon atoms are held together by strong covalent bonds

- **Phosphorous, sulfur, chlorine** and **argon** are non-metallic elements
  - Phosphorous, sulfur and chlorine exist as simple molecules (P_4, S_8, Cl_2)
  - Argon exists as single atoms
- The **covalent bonds within** the molecules are strong, however, **between** the molecules there are only weak **instantaneous dipole-induced dipole forces**
- It doesn't take much energy to break these **intermolecular** forces
- The lack of delocalised electrons means that these compounds cannot conduct electricity

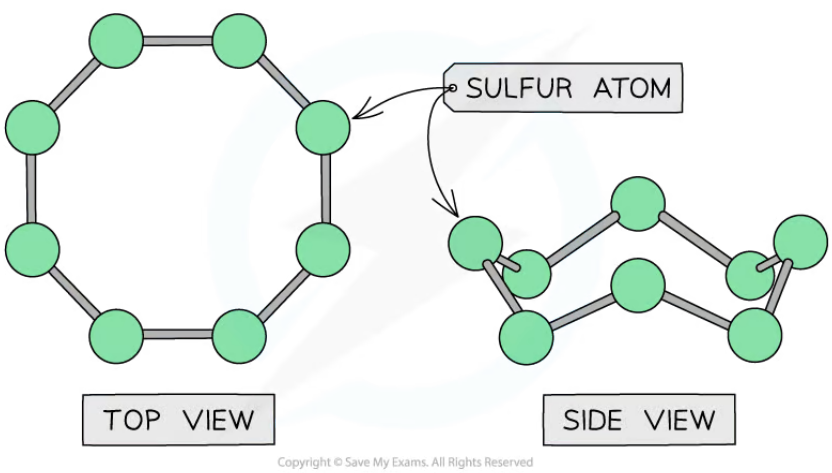
The simple molecular structure of phosphorous

Your notes



The diagram shows the simple molecular structure of phosphorus with covalent bonds between the atoms

The simple molecular structure of sulfur



The diagram shows the simple molecular structure of sulfur with covalent bonds between the atoms

Your notes

Period 3 Chlorides & Oxides

Bonding in Period 3 Chlorides & Oxides

Period 3 chlorides

- The bonding and structure of the Period 3 elements are summarised in the table below:

Bonding & structure of Period 3 elements table

Period 3 element	Na	Mg	Al	Si	P	S	Cl	Ar
Bonding	Metallic	Metallic	Metallic	Covalent	Covalent	Covalent	Covalent	-
Structure	Giant metallic	Giant metallic	Giant metallic	Giant molecular	Simple molecular	Simple molecular	Simple molecular	Simple molecular

- The table shows that **Na**, **Mg** and **Al** are metallic elements which form positive ions arranged in a **giant lattice** in which the ions are held together by a 'sea' of delocalised electrons around them
- The electrons in the 'sea' of delocalised electrons are those from the **valence shell** of the atoms
- **Na** will donate one electron into the 'sea' of delocalised electrons, **Mg** will donate two and **Al** three electrons
- As a result of this, the metallic bonding in **Al** is stronger than in **Na**
- This is because the electrostatic forces between a **3+ ion** and the larger number of negatively charged delocalised electrons is much larger compared to a **1+ ion** and the smaller number of delocalised electrons in Na
- Because of this, the **melting points increase** going from **Na** to **Al**
- **Si** has the highest melting point due to its giant molecular structure in which each Si atom is held to its neighbouring Si atoms by **strong covalent bonds**

- **P, S**, **Cl** and **Ar** are non-metallic elements and exist as **simple molecules** (P_4, S_8, Cl_2 and Ar as single atom)
- The **covalent bonds within** the molecules are strong, however **between** the molecules there are only weak **instantaneous dipole-induced dipole forces**
- It doesn't take much energy to break these **intermolecular** forces
- Therefore, the melting points decrease going from **P** to **Ar** (note that the melting point of S is higher than that of P as sulphur exists as larger S_8 molecules compared to the smaller P_4 molecule)
- The presence of a 'sea' of delocalised electrons also determines whether the element is a good conductor or not
- Going across the period the electrical conductivity of the elements decreases due to a lack of delocalised electrons

- The electronegativities of the Period 3 elements therefore determine the chemical bonding and structure of their chlorides and oxides

Chemical bonding & structure of Period 3 chlorides table

Period 3 chloride	$NaCl$	$MgCl_2$	Al_2Cl_6	$SiCl_4$	PCl_5	SCl_2
Chemical bonding	Ionic	Ionic	Covalent	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Simple molecular	Simple molecular	Simple molecular	Simple molecular

Chemical bonding & structure of Period 3 oxides table

Period 3 oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_2 SO_3
Chemical bonding	Ionic	Ionic	Ionic (with some covalent character)	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Giant ionic	Giant covalent	Simple molecular	Simple molecular

- Going across Period 3, their chlorides and oxidised become more **covalent** and their structure shifts from a **giant ionic** to a **simple molecular** structure
- Their reactions with water become more **vigorous** as a result of this as it becomes easier to **hydrolyse** the chlorides and oxides

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